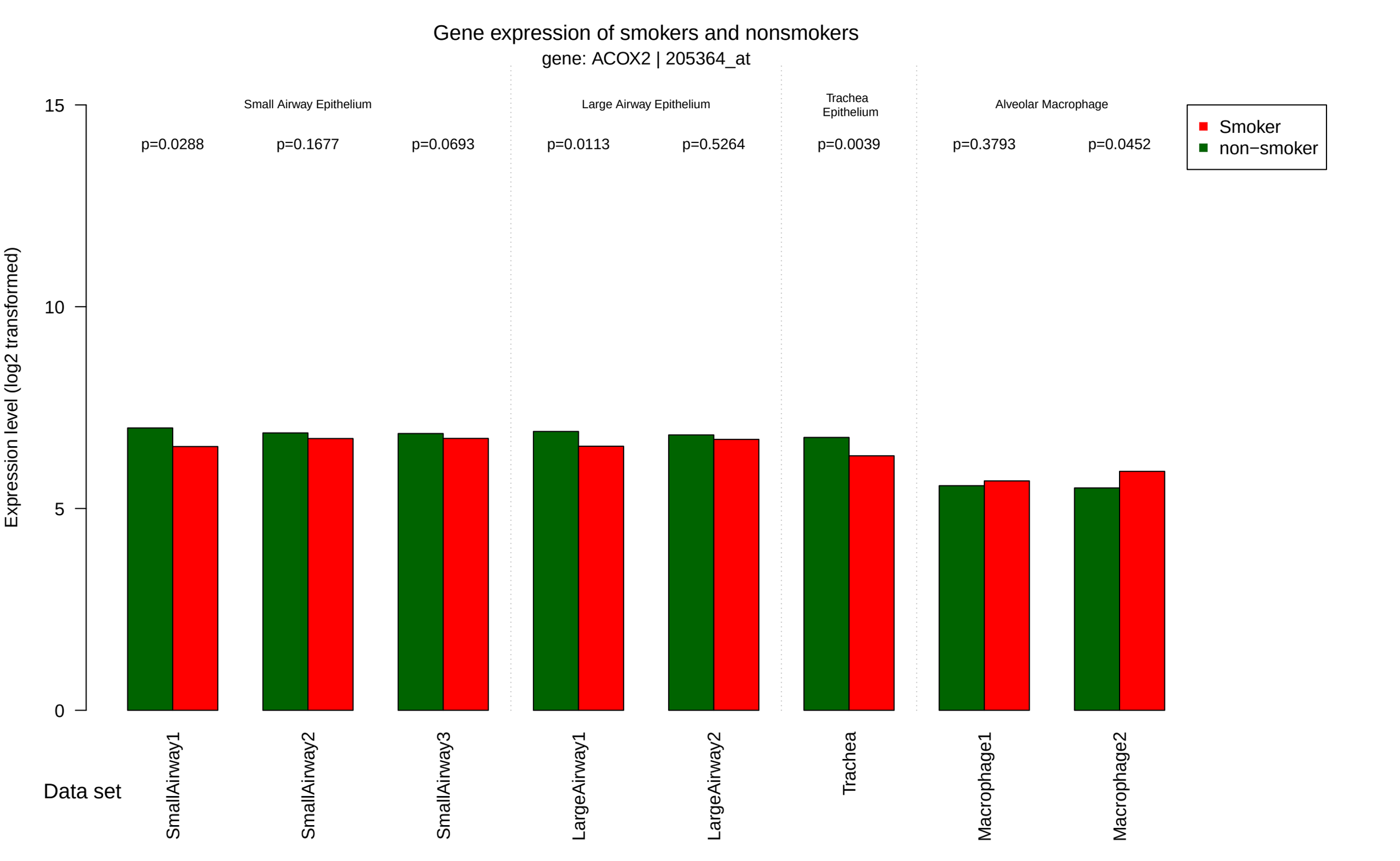
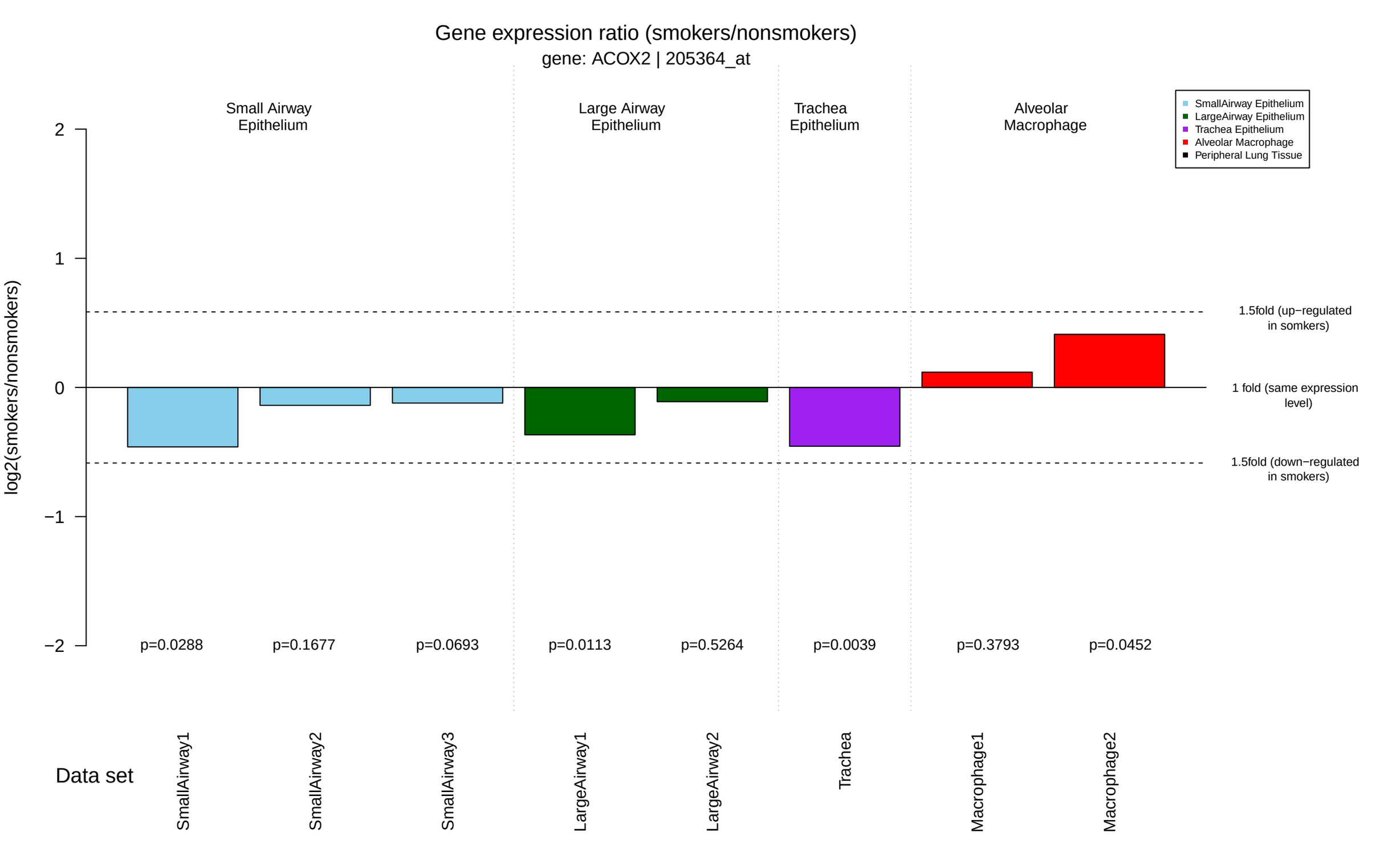
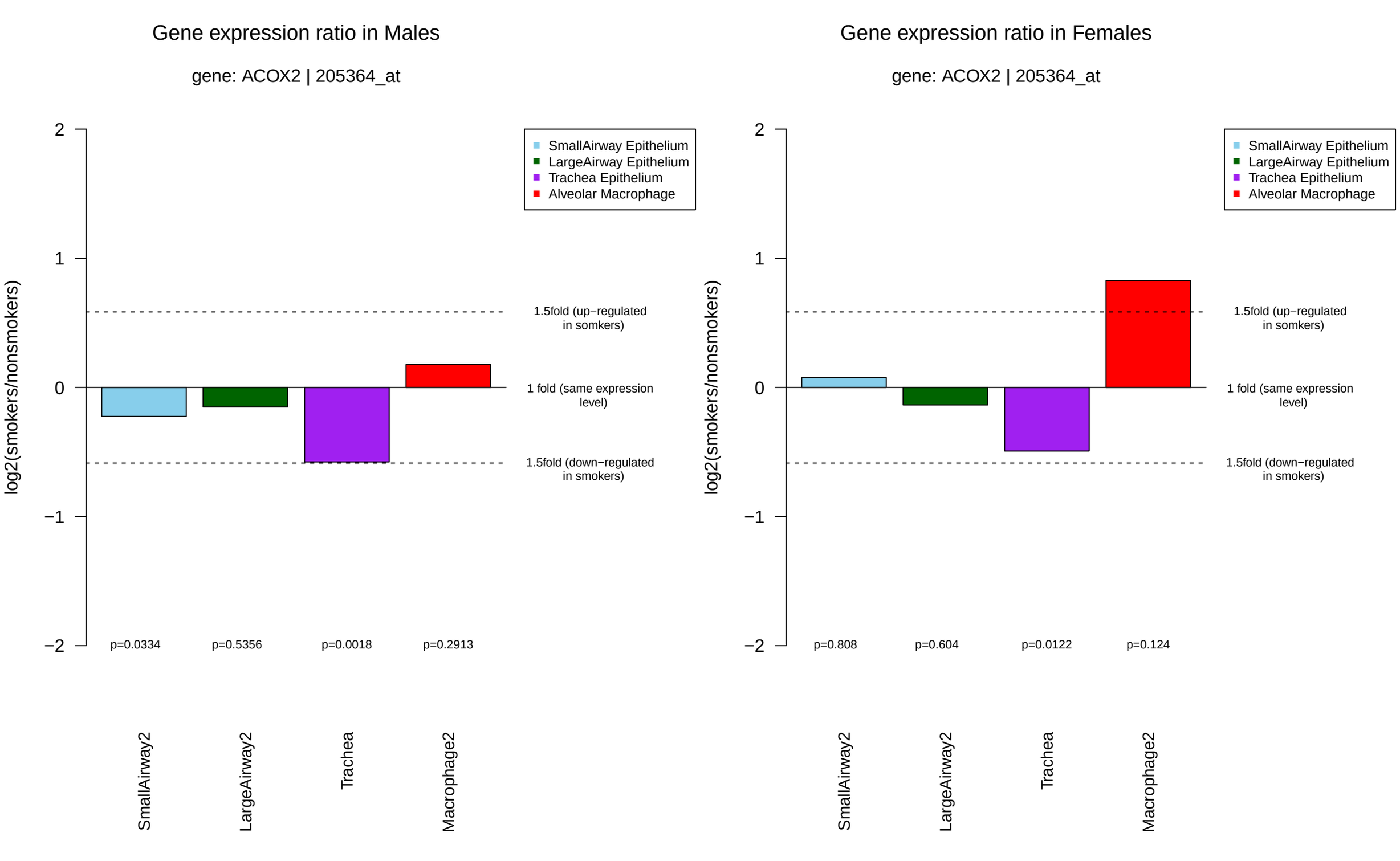
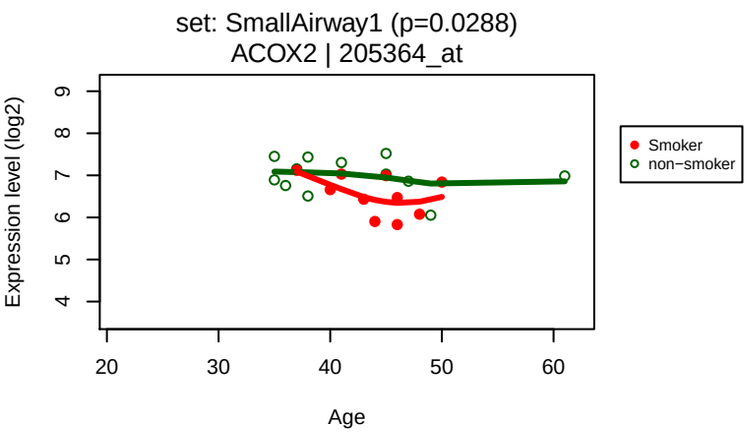
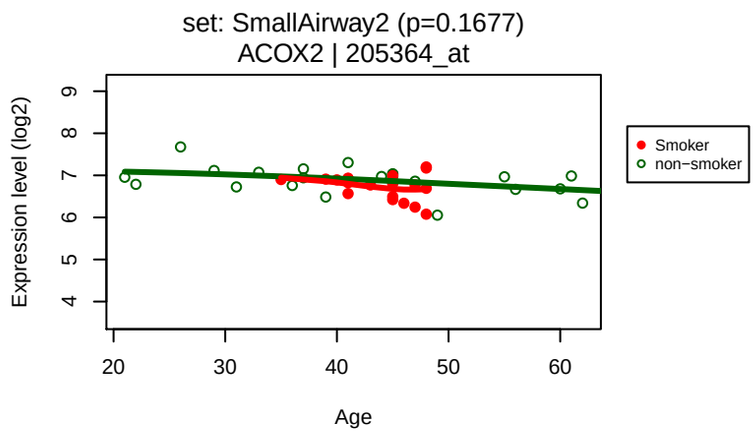
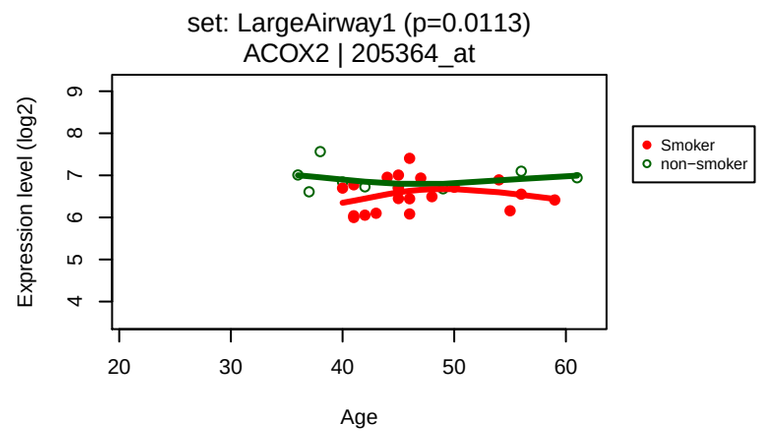

Gene expression of smokers and nonsmokers
gene: ACOX2 | 205364_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0288
p=0.1677
p=0.0693
p=0.0113
p=0.5264
p=0.0039
p=0.3793
p=0.0452
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

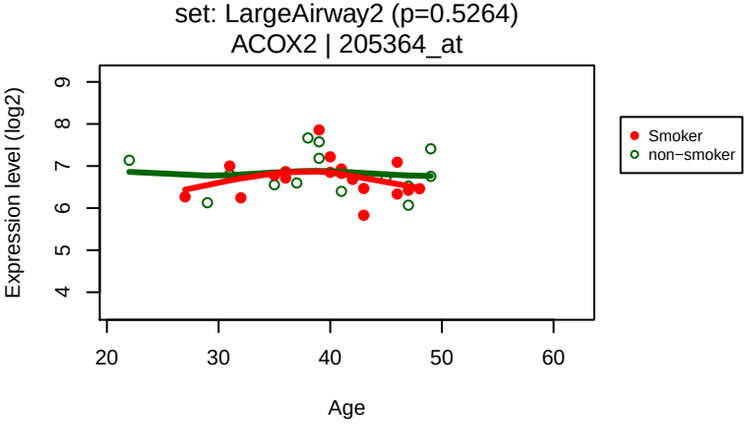

Gene expression ratio (smokers/nonsmokers)
gene: ACOX2 | 205364_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0288
p=0.1677
p=0.0693
p=0.0113
p=0.5264
p=0.0039
p=0.3793
p=0.0452
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

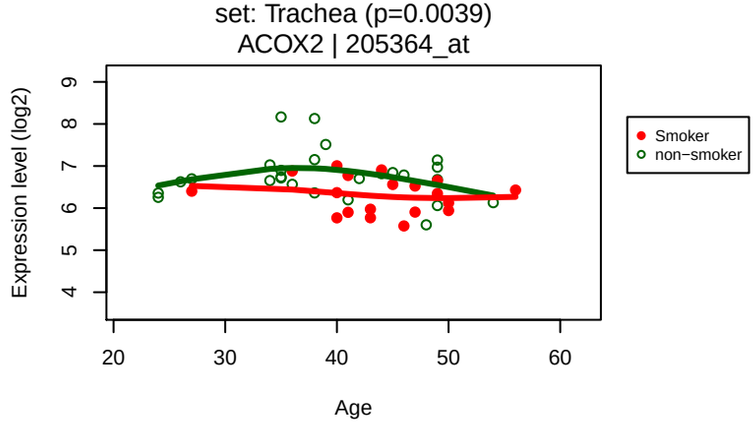

Gene expression ratio in Males
gene: ACOX2 | 205364_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0334
p=0.5356
p=0.0018
p=0.2913
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACOX2 | 205364_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.808
p=0.604
p=0.0122
p=0.124
SmallAirway2
LargeAirway2
Trachea
Macrophage2

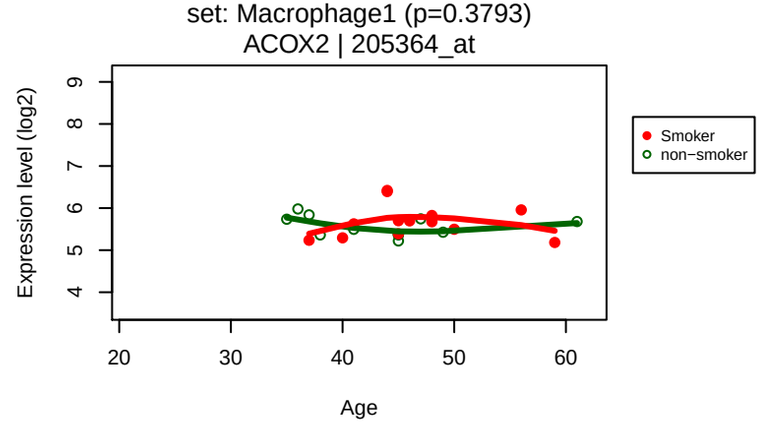

set: SmallAirway1 (p=0.0288)
ACOX2 | 205364_at
Expression level (log2)
Age
Smoker
non−smoker

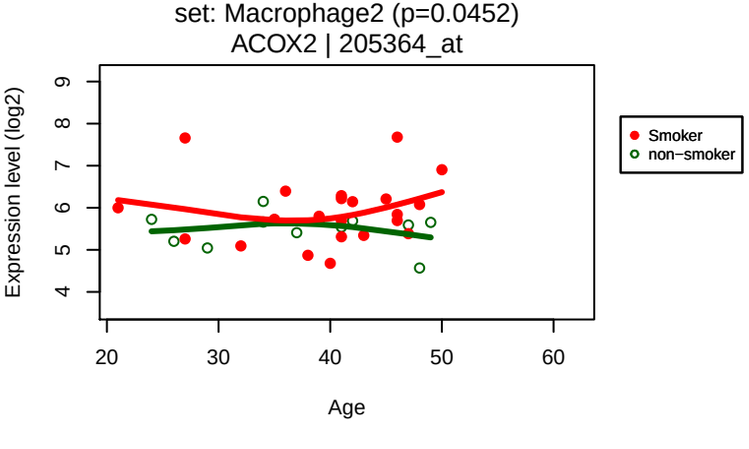

set: SmallAirway2 (p=0.1677)
ACOX2 | 205364_at
Expression level (log2)
Age
Smoker
non−smoker

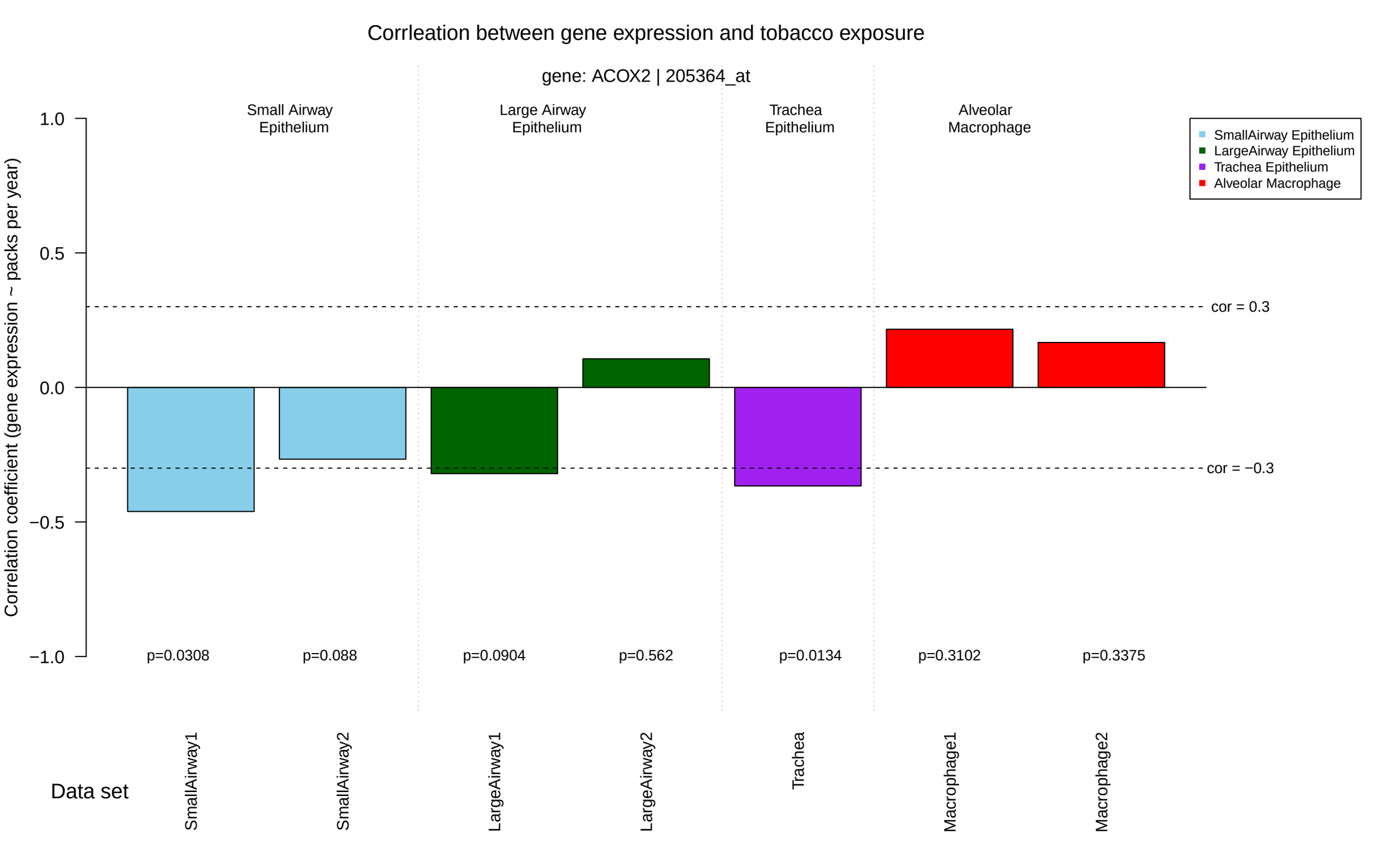

set: LargeAirway1 (p=0.0113)
ACOX2 | 205364_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5264)
ACOX2 | 205364_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0039)
ACOX2 | 205364_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3793)
ACOX2 | 205364_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0452)
ACOX2 | 205364_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACOX2 | 205364_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0308
p=0.088
p=0.0904
p=0.562
p=0.0134
p=0.3102
p=0.3375
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2